

Quiz: Recursion

Total Score: $/2^2$

Printed Name:

Quiz rules:

1. You MAY use any printed or handwritten notes.
2. You MAY NOT use a computer or any other electronic device.

Problem 1. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) == 0:
5         return 0
6     ret = foo(xs[1:])
7     return xs[0] + ret * 2
8 try:
9     xs = [1, 2, 3, 4, 5]
10    print('foo(xs)=',foo(xs))
11 except RuntimeError:
12    print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: $16 / 16 = 1.00$

Problem 2. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     if len(xs) <= 1:
5         return 0
6     return xs[0] + foo(xs[1:]) + foo(xs[2:])
7 try:
8     print('foo([1, 2, 3])=',foo([1, 2, 3]))
9 except RuntimeError:
10    print('StackOverflow')
11 EOF
12 $ python3 foo.py
```

Fraction of LLMs with correct answer: $10 / 16 = 0.62$

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def sum_even(xs):
4     return xs[0] + sum_odd(xs[1:])
5
6 def sum_odd(xs):
7     return xs[0] + sum_even(xs[1:])
8 try:
9     xs = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]
10    print('sum_even(xs)=',sum_even(xs))
11 except RuntimeError:
12    print('StackOverflow')
13 EOF
14 $ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def binary_search(xs, y):
4     def go(lo, hi):
5         if lo >= hi:
6             return False
7         mid = (lo + hi) // 2
8         if xs[mid] > y:
9             return go(lo, mid - 1)
10        if xs[mid] < y:
11            return go(mid + 1, hi)
12        return True
13    return go(0, len(xs) - 1)
14 try:
15     xs = [1, 3, 5, 7, 9]
16     print('binary_search(xs, 9)=',binary_search(xs, 9))
17 except RuntimeError:
18     print('StackOverflow')
19 EOF
20 $ python3 foo.py
```

Fraction of LLMs with correct answer: 15 / 16 = 0.94

LLM Model Performance

